# A Gradient Framework for Connotative Meaning: Weak Entailments and Conventionalization

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**Abstract**

This paper challenges the traditional lexical-encyclopedic dichotomy in formal semantics by proposing that all linguistic meaning exists on a gradient of conventionalization rather than forming categorically distinct types of knowledge. Through an analysis of entailment strength and intersubjective agreement patterns, I argue that the field's persistent failure to theorize connotative meaning stems from systematically conflating two distinct phenomena: *cultural connotation* (systematic, community-level patterns that generate weak entailments) and *private connotation* (idiosyncratic, individual associations). Drawing on evidence from semantic prosody research, arbitrariness of the linguistic sign, language acquisition patterns, and computational modeling, I develop a gradient framework in which meaning components are distinguished by their level of conventionalization rather than by categorical membership in "linguistic" versus "encyclopedic" knowledge. This reconceptualization provides theoretical tools for understanding how lexical semantic profiles generate systematic distributional patterns while remaining resistant to conscious introspection, thereby resolving longstanding puzzles in the relationship between word meaning and usage patterns.

*Keywords:* lexical semantics, connotation, entailment, semantic prosody, conventionalization, lexical-encyclopedic boundary

**A Gradient Framework for Connotative Meaning: Weak Entailments and Conventionalization**

The relationship between what words mean and how they are used presents one of the fundamental puzzles in linguistic theory. On one hand, formal semantics has developed sophisticated frameworks for analyzing compositional meaning, truth conditions, and logical entailments. On the other hand, corpus linguists have documented systematic patterns in word usage that surprise native speakers and resist traditional semantic analysis. This tension is perhaps nowhere more evident than in research on semantic prosody, where words like *cause*, *commit*, and *regime* show consistent associations with particular semantic or evaluative contexts despite speakers' inability to consciously articulate these patterns (Louw, 1993; Stubbs, 2001).

Traditional approaches to this puzzle have relied on a categorical distinction between lexical (or linguistic) knowledge and encyclopedic (or world) knowledge. Under this view, lexical semantic properties constitute the proper domain of linguistic theory—systematic, compositional, shared across speakers—while encyclopedic associations belong to pragmatics, cultural knowledge, or individual psychology. This boundary has been invoked to explain why certain patterns are theorizable while others remain resistant to formalization. However, the boundary itself has proven remarkably unstable, with theorists disagreeing about which aspects of meaning properly belong on which side (Haiman, 1980; Jackendoff, 2002; Langacker, 1987).

I propose that this instability reflects a fundamental error in conceptualization: the lexical-encyclopedic distinction is not a categorical boundary between types of knowledge but rather a gradient of conventionalization. All linguistic meaning, I argue, is ultimately encyclopedic in nature—that is, learned, arbitrary, and contingent—with what we call "lexical" knowledge representing simply the highest degree of intersubjective agreement and conventionalization. This reconceptualization has profound implications for understanding connotative meaning, which has remained chronically undertheorized in formal semantics despite its obvious systematicity in corpus patterns.

The key diagnostic error, I argue, is the field's conflation of two distinct phenomena under the single label "connotation": *cultural connotation* (systematic, community-level patterns with moderate intersubjective agreement) and *private connotation* (idiosyncratic, individual associations with minimal agreement). Cultural connotation generates what I term *weak entailments*—systematic but defeasible inferences that pattern differently from classical logical entailments but nevertheless show predictable cross-speaker consistency. Private connotation, by contrast, consists of purely associative networks that vary across individuals and lack the systematicity required for theoretical treatment at the community level.

By failing to distinguish these phenomena, the field has treated all connotation as if it were private, thereby concluding that connotative meaning lies outside the proper scope of semantic theory. This conclusion is premature and empirically incorrect. Cultural connotation exhibits sufficient systematicity to warrant formal treatment, as evidenced by consistent corpus patterns, cross-speaker agreement on subtle semantic judgments, and the successful extraction of such patterns by computational language models.

This paper develops these arguments in several stages. First, I examine the traditional lexical-encyclopedic dichotomy and document its persistent instability across theoretical frameworks, presenting five converging lines of evidence that this boundary is better understood as a gradient than a categorical distinction. Second, I develop a formal framework for understanding meaning as graded conventionalization, operationalized through levels of intersubjective agreement. Third, I introduce the crucial distinction between strong, default, and weak entailments, arguing that entailment strength correlates with conventionalization level. Fourth, I diagnose the field's systematic conflation of cultural and private connotation and demonstrate how this has led to the persistent undertheorization of connotative meaning. Fifth, I apply this framework to semantic prosody phenomena, showing how treating prosodic patterns as cultural connotation generating weak entailments resolves several longstanding puzzles. Finally, I explore broader implications for semantic theory, language acquisition, computational modeling, and cross-linguistic semantics.

The theoretical payoff of this reconceptualization extends beyond resolving specific empirical puzzles. By providing tools for formalizing graded meaning distinctions and weak entailments, this framework enables semantic theory to engage with a wider range of systematic linguistic phenomena while maintaining theoretical rigor. Moreover, it offers a more psychologically and computationally plausible model of how meaning is represented and processed, one that aligns with contemporary evidence from acquisition studies, neurolinguistics, and machine learning.

## The Instability of the Lexical-Encyclopedic Boundary

The distinction between lexical (linguistic, semantic) knowledge and encyclopedic (world, conceptual) knowledge has played a foundational role in formal semantics since at least Katz and Fodor's (1963) pioneering work on semantic theory. The basic intuition is straightforward: linguistic knowledge comprises those aspects of meaning that are properly part of the language system—systematic, compositional, shared across speakers—while encyclopedic knowledge consists of facts about the world that speakers happen to know. Under this view, knowing that *bachelor* means "unmarried adult male" is linguistic knowledge, while knowing that bachelors typically live in apartments or eat takeout food is encyclopedic.

This distinction has been invoked to solve several theoretical problems. It provides a principled basis for limiting the scope of semantic theory, preventing the incorporation of all world knowledge into lexical entries. It offers an explanation for why certain inferences pattern like entailments while others seem more like defeasible expectations. And it aligns with intuitions about compositionality: linguistic meanings combine systematically to generate complex meanings, while encyclopedic knowledge does not obviously compose in the same way.

However, despite its theoretical utility, the lexical-encyclopedic boundary has proven remarkably difficult to maintain in practice. Attempts to draw a clear line have consistently encountered problems, with theorists disagreeing about the proper classification of numerous meaning components. Consider the following cases: Is it linguistic knowledge that *kill* means "cause to die"? Most would say yes. But is it linguistic knowledge that killing typically involves force or violence? That killing is generally considered wrong? That accidental killings differ morally from intentional ones? Different theories give different answers, and the boundary becomes increasingly unclear as we move from core definitional features to more peripheral but still systematic associations.

Similar problems arise with color terms. Knowing that *red* refers to a particular region of the color spectrum seems clearly linguistic. But what about knowing that red is associated with danger, passion, or heat in many cultures? These associations show remarkable consistency across speakers and influence actual language use in systematic ways, yet they are typically classified as encyclopedic rather than linguistic. The problem is not merely one of borderline cases. Rather, sustained investigation suggests that the boundary may not be a natural one.

**Five Converging Lines of Evidence for a Gradient View**

*Theoretical Disagreements Across Frameworks.* The boundary is persistently contested across different theoretical frameworks. Generative semanticists in the 1970s argued for incorporating more world knowledge into semantic representations (Lakoff, 1971), while formal semanticists sought to maintain a stricter boundary (Katz, 1977). Cognitive linguists have largely rejected the distinction altogether, arguing that all meaning is essentially encyclopedic (Croft &

Cruse, 2004; Langacker, 1987). Frame semantics acknowledges that understanding even basic word meanings requires access to structured background knowledge (Fillmore, 1982). These persistent theoretical disagreements suggest that the boundary may not be a robust empirical phenomenon but rather a theoretical construct whose utility varies across frameworks.

*Psycholinguistic Processing Evidence.* Psycholinguistic evidence fails to support a categorical processing distinction. If lexical and encyclopedic knowledge were truly distinct types, we might expect to see different processing signatures, separate neural substrates, or differential vulnerability to brain damage. While there is evidence for some specialization in language processing, the picture that emerges is one of graded differences and extensive interaction rather than categorical separation (Federmeier & Kutas, 1999; Hagoort et al., 2004). Semantic priming studies show that both "core" semantic features and "peripheral" encyclopedic associations can facilitate processing, with the strength of priming effects varying continuously rather than categorically.

*Cross-Linguistic Variation Patterns.* The distinction becomes particularly problematic when examining cross-linguistic variation. Languages differ not only in which concepts they lexicalize but also in how they organize semantic space. The classic example involves color terms: while all languages make color distinctions, the specific boundaries and number of basic color categories vary systematically across languages (Berlin & Kay, 1969). If color category boundaries were part of universal conceptual structure (encyclopedic), we would expect more uniformity. If they were purely linguistic, we would expect more arbitrary variation. The actual pattern—systematic variation constrained by perceptual and cognitive factors—suggests a middle ground that the categorical distinction fails to capture.

*Language Acquisition Mechanisms.* Language acquisition patterns do not support learning mechanisms that would differentially acquire lexical versus encyclopedic knowledge.

Children learn word meanings through the same evidence that they learn facts about the world: observing language use in context and extracting statistical patterns. They do not appear to have separate learning mechanisms for acquiring "linguistic" meanings versus "world" knowledge (Clark, 2009; Tomasello, 2003). The gradual acquisition of word meanings, with increasing precision and refinement over time, suggests a continuous process of hypothesis formation and revision rather than the discovery of discrete categorical boundaries.

*Computational Language Models.* The success of computational language models challenges the necessity of the distinction. Large language models trained purely on distributional patterns in text acquire representations that capture both what traditional theory calls "lexical" meaning and what it calls "encyclopedic" knowledge (Brown et al., 2020; Devlin et al., 2019). These models make no architectural distinction between types of meaning; rather, they extract statistical patterns at multiple levels of abstraction from the same training signal. The fact that such models can achieve human-like performance on semantic tasks suggests that the lexical-encyclopedic distinction may not be necessary for representing or processing meaning.

These convergent lines of evidence suggest that the lexical-encyclopedic boundary is not a natural division in the structure of meaning but rather an artifact of particular theoretical commitments. If the boundary is not natural, we need an alternative framework for understanding how different aspects of meaning relate to one another and why some aspects seem more "core" or "linguistic" than others. The next section develops such a framework based on the concept of graded conventionalization.

## Meaning as Graded Conventionalization

I propose that what distinguishes different aspects of meaning is not their membership in categorically distinct types of knowledge but rather their level of conventionalization, operationalized as the degree of intersubjective agreement among speakers. *Intersubjective agreement* refers to the extent to which members of a speech community share the same associations, inferences, or judgments about a word's meaning. This reconceptualization treats meaning as existing on a continuous gradient rather than in discrete categories.

### The Foundation: Arbitrariness of Linguistic Signs

The foundation of this proposal rests on a fundamental property of linguistic signs: their arbitrariness. As Saussure (1916/1959) emphasized, the relationship between a word and its meaning is conventional rather than natural. There is no intrinsic reason why the sound sequence /dɔg/ should refer to canines in English; this association is purely a matter of social convention. Importantly, this arbitrariness extends beyond the sound-meaning pairing to all aspects of semantic structure. The fact that English *bachelor* encodes both "unmarried" and "adult male" as definitional features while leaving marital desirability or apartment-dwelling as peripheral is itself a matter of convention. Different languages make different choices about which features to encode lexically and which to leave to inference.

If all aspects of word meaning are ultimately conventional—learned through exposure to language use in specific cultural and social contexts—then what we call "linguistic" or "lexical" knowledge simply represents the highest levels of conventionalization: those aspects of meaning that show near-universal agreement among speakers of a language community. Conversely, what we call "encyclopedic" or "world" knowledge represents lower levels of conventionalization: aspects of meaning that show more variation across speakers, subgroups, or contexts.

### A Formal Framework for Gradient Conventionalization

This proposal can be formalized through a gradient framework based on levels of intersubjective agreement. Intersubjective agreement can be operationalized through multiple converging measures: explicit judgment tasks where speakers rate feature-word associations, corpus-based measures of distributional consistency across diverse text types, experimental priming studies measuring facilitation strength, and computational measures of feature activation in language models trained on naturalistic data. The convergence across these measures provides robust evidence for conventionalization levels.

Let us define an agreement score $\alpha$ as ranging from 0 (no agreement) to 1 (universal agreement) on the association between a semantic feature and a word. We can then identify several reference points along this gradient, though these represent convenient analytical categories rather than natural boundaries. At the highest end ($\alpha$ approaching 1.0), we find *reference*: the basic extensional mapping that defines what entities a word applies to. For instance, virtually all English speakers agree that *dog* applies to four-legged canine mammals.

Slightly lower ($\alpha > 0.90$) we find *denotation*: core definitional features showing very high agreement. The inference from *bachelor* to "unmarried adult male" represents this level. At a moderate-high level ($\alpha$ = 0.70-0.90), *sense relations* capture how meaning relates to other words in the semantic field, such as how *bachelor* contrasts with *married man* but aligns with *eligible*.

At a moderate level ($\alpha$ = 0.50-0.70), we find *cultural connotation*: systematic evaluative or stylistic associations showing moderate agreement. The tendency for *regime* to appear in contexts involving authoritarian or oppressive governance occupies this range. Finally, at low levels ($\alpha < 0.50$), *private connotation* reflects individual experiences with minimal cross-speaker agreement, such as one person's association of *dog* with childhood memories.

Several features of this framework warrant emphasis. First, the gradient is continuous rather than consisting of discrete levels. The categories above represent convenient reference points for theoretical discussion rather than natural boundaries. Second, agreement levels need not decline uniformly; they may drop off more steeply at certain points along the gradient. Third, different meaning components of the same word may occupy different positions on the gradient.

While the basic denotation of *bachelor* shows very high agreement, its cultural connotations (associations with particular lifestyles or social positions) show much more variation across speakers and communities.

This framework generates several empirical predictions. If meaning is gradient rather than categorical, we should observe continuous variation in agreement scores rather than clustering at discrete points. We should find correlation between agreement level and processing measures such as reaction time and priming strength. Language acquisition should show gradient patterns, with higher-agreement features learned earlier and more robustly. Cross-linguistic variation should affect which features reach which conventionalization levels for translation-equivalent words. Individual variation in agreement should pattern with demographic and experiential factors.

Preliminary evidence supports these predictions. Psycholinguistic studies using feature verification tasks show continuous variation in response times and error rates across different semantic features, rather than a categorical split (McRae et al., 2005). Developmental studies document gradual refinement of word meanings rather than sudden acquisition of categorical boundaries (Carey, 2009). Cross-linguistic research demonstrates that features encoded as "core" meaning in one language may be peripheral or pragmatic in another (Malt et al., 1999). These patterns align with a gradient view of meaning but remain puzzling under categorical frameworks.

The gradient framework provides a natural explanation for the persistent instability of the lexical-encyclopedic boundary. Theorists disagree about the boundary precisely because they are attempting to draw categorical lines through continuous space. Different theoretical frameworks effectively place the boundary at different points along the gradient, with more restrictive theories requiring very high agreement for inclusion in lexical semantics and more inclusive theories accepting moderately conventionalized features as linguistic. The framework also explains a puzzling asymmetry in traditional theory: why the boundary seems clear in prototypical cases but becomes increasingly problematic as we examine borderline cases. This pattern is exactly what we would expect if we were imposing categorical boundaries on a continuous gradient. The center of the distribution (very high and very low agreement) appears categorical because most items cluster at the extremes, while the middle range appears problematic because it genuinely represents intermediate degrees of conventionalization.

## A Theory of Weak Entailments

The reconceptualization of meaning as graded conventionalization requires a parallel reconceptualization of how meanings generate inferences. Traditional semantic theory focuses primarily on strong entailments: inferences that follow necessarily from meaning and cannot be cancelled without contradiction. However, if meaning exists on a gradient, we should expect entailments to exist on a corresponding gradient of strength. This section develops a typology of entailments distinguished by their conventionalization levels and cancellability properties.

### Beyond Classical Entailment

An entailment, in the classical sense, is a relation between propositions such that whenever the first proposition is true, the second must also be true. For example, "John is a bachelor" entails "John is unmarried"; it is impossible for the first to be true while the second is false. This notion of entailment plays a foundational role in formal semantics, serving as a diagnostic for semantic content: if a proposition $p$ entails a proposition $q$, then the information expressed by $q$ is part of the meaning of $p$.

However, many systematic patterns in language use do not fit this classical pattern. Consider the verb *cause*. Corpus analysis reveals that *cause* appears predominantly in contexts describing negative or undesirable outcomes (Stubbs, 2001). Yet it is perfectly acceptable to say

"The policy caused economic growth" or "The treatment caused a full recovery." The association with negativity is systematic and influences speakers' expectations, but it does not constitute a classical entailment because it can be overridden without contradiction.

Traditional approaches handle such cases by relegating them to pragmatics: the negative association is not part of *cause*'s semantic meaning but rather a conversational implicature or statistical generalization about usage patterns. However, this move fails to capture the systematicity of the pattern. Pragmatic implicatures should be highly context-dependent and easily cancelled, yet the prosodic patterns associated with words like *cause* are remarkably stable across corpora and contexts. We need a richer typology of entailments that recognizes gradations in entailmental strength corresponding to gradations in conventionalization.

**Three Types of Entailments**

*Strong Entailments.* Strong entailments correspond to the classical notion: inferences that follow necessarily from meaning and cannot be cancelled without contradiction. These arise from the highest levels of conventionalization ($\alpha > 0.90$). Examples include: *bachelor* → unmarried, *kill* → cause to die, *dog* → animal. Strong entailments are characterized by near-universal speaker agreement, stability across contexts, and resistance to cancellation. Denying a strong entailment results in contradiction or zeugma: "John is a bachelor but he is married" is contradictory unless *bachelor* is being used in a different sense (such as "bachelor's degree").

*Default Entailments.* Default entailments are systematic inferences that normally follow from meaning but can be overridden by explicit contextual information. These arise from high but not maximal conventionalization ($\alpha = 0.70\text{-}0.90$). Examples include: *bird* → can fly, *open* (door) → create opening by moving door on hinges. Default entailments show high speaker agreement but allow for exceptions. Saying "The emu is a bird but cannot fly" is perfectly acceptable; the default inference is cancelled by the explicit statement. However, the default still influences processing: encountering *bird* in a neutral context primes flying-related concepts more strongly than non-flying concepts (McRae et al., 1997).

*Weak Entailments.* Weak entailments are systematic but easily overridden inferences arising from moderate conventionalization ($\alpha$ = 0.50-0.70). These represent the theoretical innovation required for understanding cultural connotation. Examples include: *regime* → authoritarian or oppressive governance, *cause* → negative outcome, *utterly* → complete negative state. Weak entailments are characterized by moderate speaker agreement, high context sensitivity, and easy cancellation. They do not produce contradiction when denied but rather simply mark the usage as somewhat unexpected or requiring special context. Phrases like "democratic regime" or "cause of celebration" are perfectly acceptable, but speakers may perceive them as slightly marked or requiring explanation.

The crucial distinction between weak entailments and mere associations lies in their systematicity and cross-speaker consistency. Private associations (such as "*dog* reminds me of my childhood pet") show minimal intersubjective agreement and no systematic patterns across speakers. Weak entailments, by contrast, show moderate but consistent agreement and generate predictable patterns in corpus data, experimental measures, and computational modeling. This systematicity makes them amenable to theoretical treatment despite their defeasibility.

**The Mechanism Linking Conventionalization to Entailment Strength**

Why do higher levels of conventionalization generate stronger entailments? The answer lies in how conventionalization affects semantic representation and processing. When a particular semantic feature or inference becomes highly conventionalized—when the vast majority of speakers consistently associate it with a word—it becomes integrated into the core representation of that word's meaning. This integration occurs through repeated exposure to usage patterns across diverse contexts. As speakers encounter consistent associations between a word and certain features, these features become activated more automatically and reliably when the word is processed.

Table 1 summarizes the relationship between conventionalization level, entailment type, and key diagnostic properties.

Table 1: Conventionalization Levels and Entailment Types

| Level | Agreement Score | Entailment Type | Cancellability | Example |
|---|---|---|---|---|
| Reference | $x$ | Strong | Contradictory | *dog* → animal |
| Denotation | $x-n$ | Strong | Contradictory | *bachelor* → unmarried |
| Sense | $x-2n$ | Default | Marked but acceptable | *bird* → flies |
| Cultural | $x-3n$ | Weak | Easily overridden | *regime* → oppressive |
| Private | $x-4n$ | None | N/A | *dog* → childhood memory |

At the highest levels of conventionalization, this integration becomes so complete that the inference is no longer cognitively separable from the word's meaning. Denying the inference requires either rejecting the word's applicability altogether or interpreting the word as having a different meaning. For instance, denying that a bachelor is unmarried forces us to either reject the claim that the person is a bachelor or to interpret *bachelor* in a non-standard sense (perhaps "bachelor's degree" or an honorific title). The feature has become constitutive of the word's core meaning.

At lower levels of conventionalization, the feature remains partially separable from core meaning. Speakers recognize the association as typical or expected but do not treat it as definitional. The inference can be overridden when contextual information provides sufficient motivation, but in the absence of such information, the default expectation prevails. The processing system has learned a probabilistic association: encountering the word typically predicts the feature, but the prediction can be updated based on context.

This account predicts a correlation between agreement level and cancellability: the higher the level of intersubjective agreement on an inference, the more difficult it should be to cancel that inference without producing contradiction or marked usage. Preliminary experimental evidence supports this prediction. Features rated as more "central" or "definitional" by speakers are also more difficult to deny without perceived contradiction (Hampton, 1995). Moreover, processing studies show that violations of higher-agreement features produce stronger N400 effects (indicating semantic anomaly detection) than violations of lower-agreement features (Federmeier & Kutas, 1999).

The gradient nature of entailment strength also explains puzzling intermediate cases that have troubled traditional theory. Consider the inference from *bachelor* to "never been married." This appears weaker than the inference to "unmarried" (divorced men are unmarried but not bachelors in the prototypical sense) but stronger than mere statistical association. Under the gradient framework, this simply reflects an intermediate position on the conventionalization gradient: higher than cultural connotation but lower than core definitional features. The feature shows high but not universal agreement among speakers, creating a default rather than strong entailment.

**Cultural Versus Private Connotation**

The distinction between weak entailments and mere associations corresponds to a crucial but neglected distinction between two types of phenomena that have been conflated under the label "connotation": cultural connotation and private connotation. Failure to distinguish these has led to systematic undertheorization of connotative meaning in formal semantics. This section diagnoses the conflation problem and demonstrates the theoretically significant differences between these phenomena.

**The Conflation Problem and Its Consequences**

Traditional discussions of connotation in linguistics often treat it as a unitary phenomenon comprising subjective, affective, or evaluative associations that vary across speakers and contexts. Lyons (1977) characterizes connotative meaning as the "communicative value that an expression has by virtue of what it refers to, over and above its purely conceptual content" (p. 176). This characterization lumps together several distinct phenomena: systematic tendencies for words to appear in particular semantic or evaluative contexts (e.g., *regime* in authoritarian contexts), social or register-based associations (e.g., *father* as formal versus *dad* as casual), subjective emotional responses (e.g., *moist* as unpleasant for some speakers), and individual autobiographical associations (e.g., *beach* evoking childhood vacations).

While all of these phenomena involve associations beyond core denotational meaning, they differ fundamentally in their level of systematicity and intersubjective agreement. The first two types show considerable cross-speaker consistency and generate predictable patterns in corpus data. The latter two show much more individual variation and lack systematic distributional signatures. This conflation has had serious consequences for semantic theory. Observing that connotative associations vary across speakers and contexts, theorists have concluded that connotation lies outside the proper domain of semantic theory and belongs instead to pragmatics, sociolinguistics, or individual psychology. This conclusion would be warranted for purely private associations but is premature for systematic, community-level patterns.

**Cultural Connotation: Systematic and Theorizable**

Cultural connotation comprises associations that show moderate to high intersubjective agreement within a speech community ($\alpha$ = 0.50-0.70). These associations arise through conventional patterns of language use and are learned through exposure to how words are actually employed in discourse. Crucially, cultural connotations generate weak entailments: they lead to systematic expectations that influence processing, production, and interpretation, even if these expectations can be overridden by explicit contextual information.

Consider the example of *regime*. Corpus analysis across multiple varieties of English consistently shows that *regime* appears predominantly in contexts describing authoritarian, oppressive, or otherwise negatively evaluated systems of governance (Hunston, 2007). While *regime* can be used neutrally (as in "exercise regime" or "dietary regime"), these uses are marked and typically require contextual support. Native speakers, when asked to judge the naturalness of phrases like "democratic regime" versus "authoritarian regime," show systematic preferences that align with corpus frequencies, even if they cannot consciously articulate the association.

This pattern exhibits several key properties that distinguish cultural connotation from both strong semantic content and private association. First, *cross-speaker consistency*: The negative prosody is not idiosyncratic but shows up consistently across speakers and corpora. Second, *systematicity*: The pattern is not random but reflects structured semantic features (authoritarianism, oppression) rather than arbitrary co-occurrence. Third, *processing effects*: The association influences online processing, as demonstrated by priming studies and reading time measures. Fourth, *gradedness*: The strength of the association varies continuously with context rather than being categorical. Fifth, *cultural variation*: While systematic within a speech community, the specific associations may vary across different language communities or historical periods.

The mechanism underlying cultural connotation is straightforward: through repeated exposure to patterns of usage, speakers extract statistical regularities about the contexts in which words typically appear. These regularities become conventionalized to varying degrees, with higher-frequency patterns achieving higher levels of intersubjective agreement. The resulting expectations function as weak entailments, generating default inferences that can be overridden but nevertheless influence language processing and production. Unlike strong entailments derived from definitional features, cultural connotations reflect distributional learning from actual usage patterns.

**Private Connotation: Idiosyncratic and Non-Theorizable**

Private connotation, by contrast, comprises associations arising from individual experience and showing minimal intersubjective agreement ($\alpha < 0.50$). These associations are genuinely idiosyncratic, varying dramatically across individuals based on personal history, emotional experiences, and autobiographical memory. For example, the word *hospital* might evoke positive associations for one speaker (remembering the birth of a child), negative associations for another (remembering a parent's illness), and neutral associations for a third. These associations may be emotionally powerful for the individuals who hold them, but they show no systematic pattern across speakers and generate no predictable distributional signatures in corpus data.

Private connotations do not generate entailments in any meaningful sense. They do not create expectations about how words will be used or understood in discourse. They do not influence processing in ways that generalize across speakers. And they cannot be learned from distributional patterns in language use because they reflect extra-linguistic experience rather than conventional patterns of usage. A computational model trained on corpus data would not acquire the association between *hospital* and any particular individual's autobiographical memories, because these associations are not reflected in the distributional patterns of the corpus.

Importantly, the distinction between cultural and private connotation is not categorical but gradient. Some associations show intermediate levels of agreement—perhaps shared within subgroups but not across the entire speech community. For instance, certain professional jargon or regional dialects may show moderate agreement within specific communities but low agreement in the broader population. The key diagnostic is systematicity: do the associations generate predictable patterns in language use, or are they purely idiosyncratic? Cultural connotations, even when specific to subgroups, still show systematic patterns within those groups and can be learned from exposure to the group's linguistic practices.

**Why the Distinction Matters for Semantic Theory**

The failure to distinguish cultural from private connotation has led to a systematic undertheorization of connotative meaning in formal semantics. Observing that connotative associations vary across speakers and lack the categorical character of strong entailments, theorists have concluded that all connotation lies outside semantic theory's proper scope. This conclusion rests on the implicit assumption that all connotation is private. Once we recognize that cultural connotation exhibits sufficient systematicity to warrant theoretical treatment, new research questions emerge that have been largely neglected because the phenomenon they address has been misclassified.

First, we can investigate the relationship between cultural connotation and other types of non-truth-conditional meaning such as presupposition and conventional implicature. All three phenomena contribute to meaning without affecting truth conditions, but they differ in their conventionalization levels and cancellability properties. Understanding these relationships could lead to a unified theory of non-truth-conditional meaning.

Second, we can examine how cultural connotations interact with compositional semantics. Do weak entailments compose systematically, and if so, according to what principles? How do cultural connotations project through complex syntactic structures? These questions parallel longstanding debates about presupposition projection but have not been systematically explored for connotative meaning.

Third, we can study the acquisition mechanisms for cultural versus definitional features. Do children acquire cultural connotations through the same learning mechanisms as core denotational features, or do different types of input and processing support different types of meaning? Understanding acquisition could inform theories about the cognitive representation of graded meaning.

Fourth, we can analyze how cultural connotations change over time through processes of semantic change. The amelioration of *nice* from "foolish" to "pleasant" and the pejoration of *regime* from "system of government" to "authoritarian system" likely involved shifts in conventionalization levels. Tracking these changes systematically could reveal general principles of semantic evolution.

Finally, we can develop formal frameworks for representing weak entailments and gradations in entailmental strength. This requires extending current semantic formalisms to handle probabilistic or graded inferences while maintaining compositionality and explanatory power. Progress on this front would significantly expand the empirical coverage of semantic theory.

**Semantic Prosody Reconsidered**

The framework developed above provides new tools for understanding semantic prosody, a phenomenon that has resisted satisfactory theoretical treatment despite extensive corpus documentation. Semantic prosody refers to the tendency for certain words to appear consistently in contexts with particular semantic or evaluative characteristics, even when these characteristics are not part of the word's core denotational meaning (Louw, 1993; Sinclair, 1991; Stubbs, 2001). This section demonstrates how reconceptualizing prosody as cultural connotation resolves several longstanding theoretical puzzles.

**The Semantic Prosody Puzzle**

Consider again the verb *cause*. Corpus analysis across diverse text types consistently reveals that *cause* appears predominantly with complement clauses describing negative or undesirable outcomes. Stubbs (2001) documents this pattern extensively: *cause* combines readily with *problem*, *damage*, *concern*, *cancer*, *death*, and *trouble* (negative outcomes), but rarely appears with *benefit*, *improvement*, *happiness*, or *success* (positive outcomes).

Similar patterns appear with numerous other words. The verb *commit*, for instance, shows strong prosodic affinity with crime, error, or transgression contexts (*commit murder*, *commit suicide*, *commit adultery*, *commit perjury*) but appears marked with neutral or positive objects. The intensifier *utterly* patterns with negative adjectives (*utterly destroyed*, *utterly devastated*, *utterly hopeless*) but sounds odd with positive ones.

These patterns present several puzzles for traditional semantic theory. First, *systematicity without consciousness*: The patterns are highly systematic in corpus data yet largely invisible to native speakers' conscious intuitions. Speakers cannot usually articulate these prosodic preferences when asked directly but demonstrate them consistently in production and comprehension. Second, *statistical but non-arbitrary*: The patterns reflect genuine statistical tendencies in language use, not arbitrary co-occurrence, yet they do not follow from core denotational meaning in any obvious way. Third, *resistance to classical tests*: The prosodic associations do not pattern like classical entailments. Saying "The policy caused economic growth" is not contradictory or even particularly marked, yet corpus frequencies show consistent negativity bias. Fourth, *cross-linguistic variation*: While many prosodic patterns show cross-linguistic parallels, others vary across languages, suggesting conventional rather than natural grounding.

Traditional approaches have offered two main accounts, neither fully satisfactory. Corpus linguists often treat prosody as reflecting pragmatic or cultural knowledge that gets "absorbed" into words through repeated co-occurrence (Hoey, 2005). This account struggles to explain the patterns' systematicity and psychological reality: if prosody is simply absorbed contextual association, why does it show structured semantic patterns rather than arbitrary co-occurrence? Formal semanticists typically dismiss prosody as mere statistical association without theoretical status. This move fails to engage with the phenomenon's systematicity and cross-speaker consistency.

**Prosody as Cultural Connotation Generating Weak Entailments**

The framework developed in this paper suggests a different analysis: semantic prosody reflects cultural connotation operating at moderate conventionalization levels and generating weak entailments. Under this view, prosodic patterns arise not from absorption of contextual associations but from systematic encoding of semantic features that constrain distributional possibilities through their interaction with contextual semantics.

Consider *cause* more carefully. The verb's prosodic association with negativity reflects its encoding of agentive, forceful imposition—features that naturally align with imposing unwelcome states but create semantic tension with states that subjects typically welcome. When we say "The storm caused damage," the semantics of *cause* (forceful imposition) aligns naturally with the semantics of *damage* (unwelcome state). When we say "The policy caused prosperity," the same features of *cause* create tension with the welcoming nature of *prosperity*. The usage is acceptable because the entailment is weak and can be overridden, but the tension is reflected in lower corpus frequencies.

This analysis receives support from several sources. First, *entailment patterns*: Prosodic features behave like weak entailments. They create default expectations that influence processing but can be overridden by explicit contextual information. This aligns with the moderate conventionalization level ($\alpha$ = 0.50-0.70) proposed for cultural connotation. Second, *cross-speaker consistency*: Experimental studies find moderate agreement across speakers on prosodic judgments, higher than would be expected for pure statistical association but lower than for definitional features. Third, *systematicity*: Prosodic patterns reflect structured semantic features (agentivity, volitionality, evaluation) rather than arbitrary co-occurrence. The features interact with contextual semantics in predictable ways.

Fourth, *compositional interaction*: Prosodic features interact with compositional semantics, influencing interpretation at the sentence level. This suggests integration with semantic representation rather than purely pragmatic status. For instance, the prosodic features of *cause* interact with the evaluative properties of complement predicates to generate expectations about sentence meaning. Fifth, *language model extraction*: Neural language models successfully extract prosodic patterns from distributional data (Devlin et al., 2019), suggesting they reflect systematic features rather than noise. Models trained purely on corpus distributions develop representations that capture prosodic associations, demonstrating that these patterns are learnable from usage statistics.

**Semantic Profiles and Distributional Patterns**

The key theoretical move is recognizing that words encode *semantic profiles*: structured constellations of features that do not reduce to simple truth conditions but nevertheless systematically constrain usage possibilities. For *cause*, the profile includes agentivity (acting upon), directness (immediate connection), and forcefulness (overcoming resistance). These features make *cause* naturally compatible with contexts describing imposition of unwelcome states while creating semantic tension with contexts describing positive outcomes. The tension is not categorical—it can be overridden by context—but it is systematic and influences both production and comprehension.

Crucially, semantic profiles occupy intermediate conventionalization levels ($\alpha$ = 0.50-0.70). They show sufficient intersubjective agreement to generate systematic corpus patterns and processing effects but insufficient agreement to achieve the status of classical entailments. This explains their "intuition-resistant" character: speakers demonstrate knowledge of prosodic patterns through consistent production and comprehension behavior while struggling to consciously articulate them. The patterns exist at a level of conventionalization that influences processing automatically without requiring conscious awareness.

This analysis generates several testable predictions. First, prosodic strength should correlate with feature agreement: prosodies reflecting more conventionalized features should show stronger corpus effects and more consistent cross-speaker judgments. Second, prosodic violations should create processing cost: encountering prosodically mismatched contexts should slow comprehension and require additional processing, as measured by reading times or N400 amplitudes. Third, individual variation in prosodic sensitivity should pattern with exposure: speakers with more experience in relevant registers should show stronger prosodic effects. Fourth, cross-linguistic prosodic variation should reflect differences in which features are conventionalized at which levels for translation-equivalent words.

### Theoretical Implications of the Prosody Analysis

Reanalyzing semantic prosody as cultural connotation generating weak entailments has several important implications for semantic theory. First, it validates prosody as a legitimate object of semantic inquiry rather than a peripheral phenomenon. Prosodic patterns reflect systematic aspects of lexical meaning that interact with compositional semantics and influence interpretation. They are not merely statistical artifacts or pragmatic effects but genuine components of word meaning that happen to occupy moderate positions on the conventionalization gradient.

Second, it provides a principled explanation for prosody's apparently paradoxical properties: systematic yet intuition-resistant, statistical yet structured, meaningful yet non-truth-conditional. These properties follow naturally from intermediate conventionalization levels. The systematicity reflects genuine conventionalization above random chance. The intuition-resistance reflects insufficient conventionalization for conscious access. The statistical nature reflects the gradient and probabilistic character of weak entailments. The structured patterns reflect the semantic content of the underlying features.

Third, it suggests productive connections between prosody research and other domains of semantic theory, including presupposition, conventional implicature, and expressive meaning. All of these phenomena may reflect different types of conventionalized meaning operating at different gradient levels. Presuppositions might occupy higher conventionalization levels than prosody, while expressives might occupy similar or lower levels. Mapping these relationships systematically could lead to a unified theory of non-truth-conditional meaning organized along the conventionalization gradient.

Fourth, it opens new methodological possibilities. If prosodic features reflect moderate conventionalization, they should be measurable through agreement studies, manipulable in experimental contexts, and learnable from distributional evidence. These predictions enable empirical testing of the framework using converging evidence from corpus analysis, psycholinguistic experimentation, and computational modeling. The framework thus transforms prosody from a curious descriptive phenomenon into a theoretically significant testing ground for claims about meaning representation.

### Broader Implications and Future Directions

The gradient framework for meaning and the theory of weak entailments developed in this paper have implications extending beyond the specific phenomena discussed above. This section explores several areas where the framework may prove productive, identifying both opportunities for theoretical development and empirical challenges that remain to be addressed.

#### Implications for Formal Semantic Theory

For formal semantic theory, the primary implication is the need for representational frameworks that can capture graded distinctions in conventionalization and entailmental strength. Classical truth-conditional semantics operates with categorical distinctions: meaning components either are or are not part of a word's semantic content, and inferences either do or do not constitute entailments. This categorical approach works well for core truth-conditional content but struggles with the middle ranges of the conventionalization gradient.

Several existing frameworks provide starting points for incorporating gradience. Prototype theory (Rosch, 1975) already recognizes graded category membership, though it focuses primarily on extensional rather than intensional gradations. Frame semantics (Fillmore, 1982) allows for structured background knowledge, though it lacks explicit mechanisms for distinguishing conventionalization levels. Generative Lexicon theory (Pustejovsky, 1995) encodes contextual affordances in lexical structure, capturing some of what we are calling weak entailments. However, none of these frameworks systematically addresses the full range of issues raised by gradient conventionalization.

A fully developed framework would need to specify several key components. First, it must represent gradient meaning distinctions formally, perhaps using weighted features or probabilistic structures. Second, it must explain how gradient meanings compose: when two words with weak entailments combine, how do their weak entailments interact? Do they simply combine additively, or do they interact in more complex ways? Third, it must account for how weak entailments project through compositional structure. Do they project like presuppositions, or do they follow different projection principles?

These challenges are substantial but not insurmountable. Recent work in probabilistic semantics and Bayesian approaches to meaning provides potential tools for formalizing gradient inferences (Goodman & Lassiter, 2015). Vector space models and distributional semantics offer methods for representing graded semantic similarity (Turney & Pantel, 2010). Integrating these approaches with traditional compositional semantics could yield frameworks capable of handling the full gradient of conventionalization while maintaining theoretical rigor.

### Implications for Language Acquisition Research

For acquisition research, the gradient framework predicts that children should acquire meaning components at rates correlated with their conventionalization levels. Highly conventionalized features (those approaching $\alpha = 1.0$) should stabilize early in development, while less conventionalized features should continue to be refined throughout childhood and even into adolescence. This prediction aligns with existing evidence showing that children acquire core denotational features relatively early but continue to refine understanding of evaluative, register-based, and contextual constraints well into late childhood (Clark, 2009).

The extended acquisition period for cultural connotations likely reflects their intermediate position on the conventionalization gradient. Because these features show moderate rather than near-universal agreement, learners need more exposure to extract the relevant patterns from noisier distributions. A child hearing *regime* used to describe both authoritarian governments and neutral administrative systems must extract the predominant association with authoritarianism from statistical patterns in input, a process requiring substantial exposure.

The framework also suggests that different types of input may differentially support acquisition of different gradient levels. Core denotational features may be learnable from ostensive labeling and explicit instruction: a parent pointing to a dog and saying "That's a dog" provides direct evidence for the basic referential mapping. Cultural connotations, by contrast, may require extensive passive exposure to naturalistic usage patterns. No parent explicitly teaches that *cause* tends to combine with negative outcomes; children must extract this pattern from observing actual usage.

This differential input requirement could inform both developmental research and language pedagogy. Researchers could investigate whether acquisition of cultural connotations correlates more strongly with passive exposure (e.g., amount of reading) than with interactive instruction. Language teachers could design instruction that provides explicit attention to prosodic patterns and cultural connotations, potentially accelerating acquisition of these intermediate-conventionalization features that might otherwise require extensive naturalistic exposure.

**Implications for Computational Modeling**

The success of neural language models in acquiring human-like semantic representations provides strong support for the gradient framework. These models extract meaning representations from purely distributional evidence without any explicit encoding of the lexical-encyclopedic boundary or categorical distinctions between meaning types (Brown et al., 2020; Devlin et al., 2019). Recent work has shown that language models acquire not only core denotational features but also subtle cultural connotations and prosodic patterns (Weissweiler et al., 2022). The models' representations show graded sensitivity to features at different conventionalization levels, with representation strength roughly correlating with corpus frequency and consistency—exactly as predicted by the gradient framework.

This suggests productive future directions for computational semantics. Rather than attempting to impose categorical boundaries on learned representations, researchers could analyze the gradient structure directly, mapping how different types of semantic features emerge at different levels of abstraction in model architectures. For instance, researchers could measure agreement between model-generated feature associations and human judgments across the conventionalization gradient, testing whether models show the predicted correlation between training corpus consistency and human intersubjective agreement.

Moreover, computational models provide tools for testing theoretical predictions about weak entailments. If cultural connotations generate weak entailments that influence processing, we should observe measurable effects in model behavior: models should show graded preference for prosodically matched contexts, these preferences should correlate with corpus statistics, and the preferences should be overridable by explicit contextual information. Testing these predictions computationally could provide large-scale evidence complementing psycholinguistic experiments.

### Implications for Cross-Linguistic Semantics

The gradient framework predicts systematic cross-linguistic variation in which features reach which conventionalization levels. Languages may differ not only in which concepts they lexicalize but also in how they structure semantic space within lexical items. For example, many languages encode evidentiality (source of information) as a grammaticalized category with strong entailmental status, while English treats evidential distinctions as largely pragmatic or lexical (Aikhenvald, 2004). Under the gradient framework, this reflects different conventionalization levels for the same semantic domain: what is highly conventionalized (high $\alpha$) in one language remains less systematized in another.

Similarly, the specific cultural connotations associated with translation equivalents often differ across languages, reflecting different patterns of usage and conventional association. The English *regime* and French *régime* share core denotational meaning but may differ in their prosodic associations and cultural connotations. Understanding these differences requires attending to gradient levels of meaning rather than assuming that translation equivalents simply vary in their "peripheral" or "encyclopedic" associations. The differences may reflect systematic variation in conventionalization levels that should be modeled explicitly.

Cross-linguistic investigation could test several predictions of the gradient framework. If conventionalization is gradient and language-specific, we should find: First, translation equivalents showing similar core denotation but different cultural connotations. Second, correlation between conventionalization levels and grammaticalization patterns (more conventionalized features should grammaticalize more readily). Third, systematic differences in which types of features reach which conventionalization levels across language families. Fourth, historical relationships between conventionalization level changes and semantic change patterns.

**Implications for Semantic Change**

The gradient framework provides new tools for understanding semantic change. Traditional approaches distinguish different types of change (narrowing, broadening, amelioration, pejoration) but struggle to explain the mechanisms driving change and the intermediate stages through which meanings pass. Under the gradient framework, semantic change can be understood as gradual shifts in conventionalization levels. Features that begin as weak entailments or cultural connotations can become increasingly conventionalized over time, eventually achieving the status of definitional features. Conversely, definitional features can lose conventionalization, becoming optional or contextually specified.

For example, the amelioration of *nice* from "foolish" to "pleasant" likely passed through intermediate stages where both associations coexisted at different conventionalization levels. Initially, the "pleasant" sense may have emerged as a weak entailment in specific contexts, gradually gaining conventionalization as usage patterns shifted. The pejoration of *regime* from neutral "system of government" to negatively associated "authoritarian system" reflects increasing conventionalization of negative associations from weak cultural connotation to near-definitional status in many contexts.

Tracking these changes requires attention to gradations in conventionalization rather than categorical shifts in meaning type. Historical corpus analysis could measure how agreement levels for specific features change over time, using distributional patterns as proxies for conventionalization. Comparing these diachronic patterns across multiple cases of semantic change could reveal general principles about how conventionalization levels shift and what factors drive or constrain these shifts. Do certain types of features tend to gain conventionalization while others tend to lose it? Are there preferred pathways through the conventionalization gradient? Does the rate of change correlate with initial conventionalization level?

**Remaining Theoretical and Empirical Challenges**

Despite its explanatory potential, the gradient framework faces several challenges that require further development. First, the compositional semantics of weak entailments remains underspecified. When two words with weak entailments combine, how do their weak entailments interact? Consider "utterly cause damage": does the weak negative entailment of *utterly* combine additively with that of *cause*, or do they interact in more complex ways? Developing formal composition principles for weak entailments is essential for a complete theory.

Second, the precise mechanisms linking conventionalization to entailment strength require further articulation. While the general principle is clear—higher conventionalization produces stronger entailments—the cognitive and computational mechanisms implementing this relationship need detailed specification. What processes in semantic representation and processing create the correlation between agreement level and cancellability? How does the language processing system track and update conventionalization levels during comprehension?

Third, empirical validation through systematic experimentation is needed. While the framework aligns with existing evidence from corpus linguistics, psycholinguistics, and computational modeling, direct tests of its specific predictions remain limited. Researchers should design experiments measuring agreement levels across the conventionalization gradient, testing correlations with processing measures, acquisition patterns, and cross-linguistic variation. Such studies would provide crucial evidence for or against the framework's core claims.

Fourth, the framework's relationship to other gradient phenomena in language deserves exploration. Gradience appears in phonology (gradient phonotactic acceptability), morphology (gradient productivity), and syntax (gradient acceptability judgments). Is the conventionalization gradient in semantics an instance of a more general principle organizing linguistic knowledge? Or does semantics involve unique gradient properties? Addressing these questions could situate the current framework within broader theories of linguistic representation.

## Conclusion

This paper has argued for a fundamental reconceptualization of how we understand lexical meaning. Rather than maintaining a categorical distinction between lexical (linguistic, semantic) and encyclopedic (world, conceptual) knowledge, I have proposed that all meaning exists on a continuous gradient of conventionalization, operationalized as degrees of intersubjective agreement among speakers. What we call "lexical" meaning simply represents the highest levels of conventionalization, while what we call "encyclopedic" knowledge occupies lower positions on the same gradient.

This reconceptualization resolves several longstanding problems in semantic theory. It explains why the lexical-encyclopedic boundary has proven so persistently unstable across theoretical frameworks: theorists have been attempting to draw categorical lines through continuous space. It provides a natural account of intermediate cases that trouble categorical approaches. And it aligns with converging evidence from the arbitrariness of linguistic signs, language acquisition patterns, computational modeling success, cross-linguistic variation, and psycholinguistic processing studies.

The gradient framework requires a parallel reconceptualization of entailments. Rather than treating entailment as a categorical relation, we need a richer typology recognizing strong entailments (necessary inferences from core definitional features), default entailments (typical but overridable inferences from high-conventionalization features), and weak entailments (systematic but easily cancelled inferences from moderate-conventionalization features). Weak entailments—systematic but defeasible inferences arising from moderate conventionalization—provide the theoretical tool needed for understanding cultural connotation.

The crucial diagnostic error in prior work has been the conflation of two distinct phenomena under the single label "connotation": cultural connotation (systematic, community-level patterns generating weak entailments) and private connotation (idiosyncratic, individual associations lacking systematicity). By failing to distinguish these, the field has treated all connotation as if it were private, thereby concluding that connotative meaning lies outside semantic theory's proper scope. This conclusion is premature. Cultural connotation exhibits sufficient systematicity to warrant theoretical treatment and can be formalized through the apparatus of gradient conventionalization and weak entailments.

Application of this framework to semantic prosody demonstrates its empirical utility. Prosodic patterns, which have resisted satisfactory theoretical treatment in traditional frameworks, emerge naturally as cultural connotations generating weak entailments. The framework explains prosody's apparently paradoxical properties: systematic yet intuition-resistant, statistical yet structured, meaningful yet non-truth-conditional. These properties follow from intermediate conventionalization levels and the weak entailmental patterns they generate. Treating prosody as cultural connotation integrates this phenomenon into semantic theory while explaining its distinctive characteristics.

The theoretical implications extend broadly across semantic theory, acquisition research, computational modeling, cross-linguistic semantics, and semantic change. By providing tools for formalizing gradient distinctions in meaning and entailmental strength, the framework enables engagement with a wider range of systematic linguistic phenomena while maintaining theoretical rigor. For semantic theory, it suggests the need for frameworks capable of representing and composing gradient meanings. For acquisition research, it predicts correlation between conventionalization levels and acquisition timelines. For computational modeling, it explains why distributional learning succeeds at acquiring both core meanings and cultural connotations. For cross-linguistic semantics, it provides tools for understanding systematic variation in how languages conventionalize semantic features. For semantic change, it offers mechanisms for understanding how meanings evolve through shifts in conventionalization levels.

Much work remains. The framework requires further formalization, particularly regarding compositional semantics of weak entailments and projection principles. Empirical validation through systematic psycholinguistic experimentation is needed to test predicted correlations between conventionalization levels, agreement scores, and processing measures. Cross-linguistic investigation could reveal whether the gradient structure is universal or subject to typological variation. Exploration of how gradient meanings are represented in neural architectures could provide insights into cognitive underpinnings of semantic representation. Development of formal frameworks for representing and composing weak entailments remains a significant theoretical challenge.

Nevertheless, the fundamental insight seems robust: meaning is not categorically partitioned into distinct types but rather exists on a continuous gradient of conventionalization. Recognizing this gradient structure, and the system of weak entailments it generates, provides theoretical tools for engaging with the full richness of systematic meaning phenomena while avoiding the pitfalls of attempting to maintain unstable categorical boundaries. The persistent puzzle of how words mean and how meanings constrain usage may find resolution not in discovering the "true" boundary between types of meaning but in recognizing that no such boundary exists—only points along a continuous landscape of conventionalization. This reconceptualization opens new avenues for semantic theory while grounding its claims in empirically observable patterns of intersubjective agreement, distributional consistency, and processing effects.

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